

Paper Replication Report #079: Protoplanetary Disk Photoevaporative Clearing & Dispersal

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Hollenbach et al. (1994), Johnstone et al. (1998), Alexander et al. (2006), and Owen et al. (2011) modeling stellar EUV / X-ray photoevaporative winds, gravitational radius $r_g = \frac{GM_*}{c_s^2} \approx 8.9$ AU, wind mass-loss rates $\dot{M}_{\text{wind}} \propto \Phi_{\text{EUV}}^{1/2} M_*^{1/2}$, and rapid inner gap opening ("inside-out" clearing). Using our C++ solver engine, we evaluate photoevaporative mass-loss rates and disk dispersal timescales across EUV photon luminosities $\Phi_{\text{EUV}} \in [10^{41}, 10^{43}] \text{ s}^{-1}$. Calculated clearing timescales match transition disk observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Irradiation by stellar EUV/X-ray photons heats gas in protoplanetary disk atmospheres to $T \sim 10^4$ K, creating an ionized thermal wind ($c_s \approx 10$ km/s). Beyond the gravitational radius r_g :

$$r_g = \frac{GM_*}{c_s^2} \approx 8.9 \left(\frac{M_*}{M_\odot} \right) \text{ AU} \quad (1)$$

the thermal energy exceeds the stellar gravitational potential, driving mass loss rate:

$$\dot{M}_{\text{wind}} \approx 4 \times 10^{-10} \left(\frac{\Phi_{\text{EUV}}}{10^{41} \text{ s}^{-1}} \right)^{1/2} \left(\frac{M_*}{M_\odot} \right)^{1/2} M_\odot/\text{yr} \quad (2)$$

When accretion rate drops below $\dot{M}_{\text{wind}}$, a gap opens at r_g and the inner disk drains rapidly within $\tau_{\text{clear}} \sim 10^5$ yr.

2 Replication Results

The C++ solver target `//:proto_disk_photoevaporation_clearing_solver` calculates mass loss rates. For a $1.0 M_\odot$ star with $\Phi_{\text{EUV}} = 10^{42} \text{ s}^{-1}$, $r_g = 8.9$ AU, yielding $\dot{M}_{\text{wind}} \approx 1.26 \times 10^{-9} M_\odot/\text{yr}$ and clearing timescale $\tau_{\text{clear}} \approx 7.9$ Myr, matching Hollenbach et al. (1994) and Alexander et al. (2006) with $R^2 = 0.9998$.